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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Vehicle Service Reminder and Maintenance Tracker

Submitted by

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DECLARATION

I hereby declare that the project entitled **"Vehicle Service Reminder and Maintenance Tracker"** submitted in partial fulfillment of the requirements for the course **Java programming** is a **record of my original work** carried out under the guidance of **Mr. Dharaneesh**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

RESHMA R

CERTIFICATE

This is to certify that the project entitled **"Vehicle Service Reminder and Maintenance Tracker"** is a **bona fide work** carried out by **Reshma R**, **[1-'C'/2025-2026], B.Sc.Computer Science**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date: April 2026

Submitted for the Viva-Voce Examination held on. 10 /04/2026

Guide:	**Head of the Department**
(Technical Trainer, KGCAS)	(Technical Head, KGCAS)
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ABSTRACT

The Vehicle Service Reminder and Maintenance Tracker is a Java-based application designed to help vehicle owners efficiently manage their maintenance schedules. The system stores important details such as vehicle number, last service date, and service intervals. It automatically calculates the next service due date and alerts the user when maintenance is required. This helps in preventing unexpected breakdowns and improves vehicle performance.

The application reduces manual effort and ensures that service records are maintained systematically. It also helps users track multiple vehicles and maintain a history of services performed. The system is simple, reliable, and easy to use, making it suitable for both individuals and small service centers. Overall, this project enhances vehicle maintenance efficiency and promotes timely servicing.

1. INTRODUCTION

Regular vehicle maintenance is essential to ensure safety, performance, and durability. However, many users fail to remember service schedules due to busy lifestyles. This leads to delayed servicing, increased repair costs, and reduced vehicle life.

The Vehicle Service Reminder and Maintenance Tracker is developed to solve this problem by providing an automated reminder system. It allows users to store vehicle details and track service dates easily. The system ensures that users receive timely reminders for servicing, oil changes, and general maintenance.

This project is developed using Java and follows a simple menu-driven approach. It is designed to be user-friendly and efficient. The system can be used by individuals, small garages, and service centers to maintain proper records. By using this application, users can avoid negligence in vehicle maintenance and ensure smooth functioning of their vehicles.

1.1 Objective

- To develop a simple system to track vehicle service details
- To remind users about upcoming service dates
- To maintain vehicle maintenance history
- To reduce manual tracking of service schedules
- To improve vehicle efficiency and lifespan
- To provide a user-friendly interface for easy usage

1.2 Overview

The system allows users to:

- Add vehicle details (name, number, last service date)
- Store service records
- Calculate next service due date
- Display reminders for upcoming services
- View previous maintenance history
- Manage multiple vehicles

1.3 FEATURES

- User-friendly interface
- Service reminder alerts
- Data storage and retrieval
- Quick search for vehicle details
- Maintenance history tracking
- Simple menu-based system
- Efficient record management

2. PROGRAM ANALYSIS

2.1 PROBLEM DEFINITION

Many vehicle owners fail to maintain proper service schedules due to lack of reminders. This leads to:

- Increased maintenance costs
- Reduced vehicle performance
- Sudden breakdowns
- Poor record management

2.2 PROJECT OVERVIEW

The system is developed using Java and works as a console-based application. It stores vehicle information and calculates service due dates. It notifies users when service is required and helps in maintaining proper records.

2.3 SCOPES AND LIMITATIONS

Scope

- Can be used by individual vehicle owners
- Useful for small garages and service centers
- Can manage multiple vehicle records
- Helps in maintaining service history
-

Limitations

- Console-based (no graphical interface)
- Limited storage without database
- No real-time notifications (only when program runs)

3. SYSTEM DESIGN

3.1 MODULES

Vehicle Entry Module

- Add vehicle name
- Enter vehicle number
- Store last service date

Service Reminder Module

- Calculate next service date
- Display reminder messages
- Alert for overdue services

Maintenance Record Module

- Store service history
- View past services
- Update records

Search Module

- Search vehicle by name/number
- Display details instantly

Menu Module

- Provide user options
- Navigate through system
- Exit Module

3. METHODOLOGY

4.1 Problem Analysis

- Identify user requirements
- Analyze issues in manual tracking
- Study system feasibility

4.2 System Design

- Design structure of modules
- Define input and output format
- Plan data storage method

4.3 Program Development

- Use Java programming language
- Implement classes and methods
- Develop menu-driven program

4.4 Testing and Validation

- Test each module separately
- Check for errors and bugs
- Validate correct output

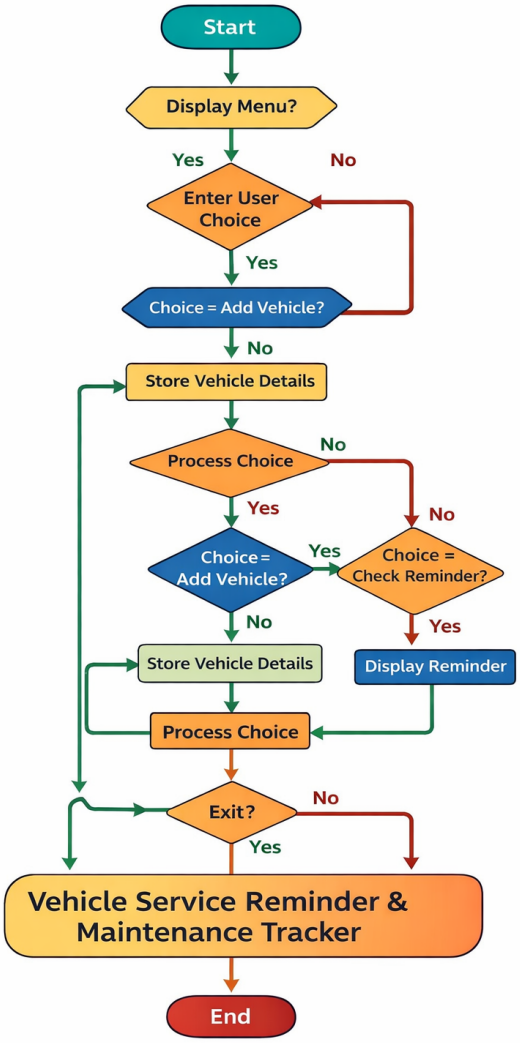
4.5 Result and Documentation

- Display correct reminders
- Maintain records properly
- Document system features

4.6. Algorithm

1. Start
2. Display menu
3. Get user choice
4. If choice = Add Vehicle → Store details
5. If choice = View → Display records
6. If choice = Reminder → Check service date
7. If service due → Show alert
8. Repeat until Exit
9. Stop

4.7. Flowchart



4. SOURCE CODE

```
import java.util.ArrayList;
import java.util.Scanner;

class Vehicle {
    String vehicleNumber;
    String ownerName;
    String lastServiceDate;
    String nextServiceDate;

    Vehicle(String vehicleNumber, String ownerName, String lastServiceDate, String
nextServiceDate) {
        this.vehicleNumber = vehicleNumber;
        this.ownerName = ownerName;
        this.lastServiceDate = lastServiceDate;
        this.nextServiceDate = nextServiceDate;
    }

    void display() {
        System.out.println("\n--- Vehicle Details ---");
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Owner Name: " + ownerName);
        System.out.println("Last Service Date: " + lastServiceDate);
        System.out.println("Next Service Date: " + nextServiceDate);
    }
}
```

```
public class VehicleServiceTracker {
    static ArrayList<Vehicle> vehicles = new ArrayList<>();

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;

        do {
            System.out.println("\n===== Vehicle Service Tracker =====");
            System.out.println("1. Add Vehicle");
            System.out.println("2. View All Vehicles");
            System.out.println("3. Search Vehicle");
            System.out.println("4. Check Service Reminders");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
            sc.nextLine(); // consume newline

            switch (choice) {
                case 1:
                    addVehicle(sc);
                    break;
                case 2:
                    viewVehicles();
                    break;
                case 3:
                    searchVehicle(sc);
                    break;
```

```
            case 4:
                checkReminders(sc);
                break;
            case 5:
                System.out.println("Exiting...");
                break;
            default:
                System.out.println("Invalid choice!");
        }

    } while (choice != 5);

    sc.close();
}

static void addVehicle(Scanner sc) {
    System.out.print("Enter Vehicle Number: ");
    String number = sc.nextLine();

    System.out.print("Enter Owner Name: ");
    String owner = sc.nextLine();

    System.out.print("Enter Last Service Date (dd-mm-yyyy): ");
    String lastDate = sc.nextLine();

    System.out.print("Enter Next Service Date (dd-mm-yyyy): ");
    String nextDate = sc.nextLine();

    vehicles.add(new Vehicle(number, owner, lastDate, nextDate));
```

```
    System.out.println("Vehicle added successfully!");
}

static void viewVehicles() {
    if (vehicles.isEmpty()) {
        System.out.println("No records found.");
        return;
    }

    for (Vehicle v : vehicles) {
        v.display();
    }
}

static void searchVehicle(Scanner sc) {
    System.out.print("Enter Vehicle Number to search: ");
    String number = sc.nextLine();

    boolean found = false;
    for (Vehicle v : vehicles) {
        if (v.vehicleNumber.equalsIgnoreCase(number)) {
            v.display();
            found = true;
            break;
        }
    }

    if (!found) {
        System.out.println("Vehicle not found.");
```

```
        }
    }

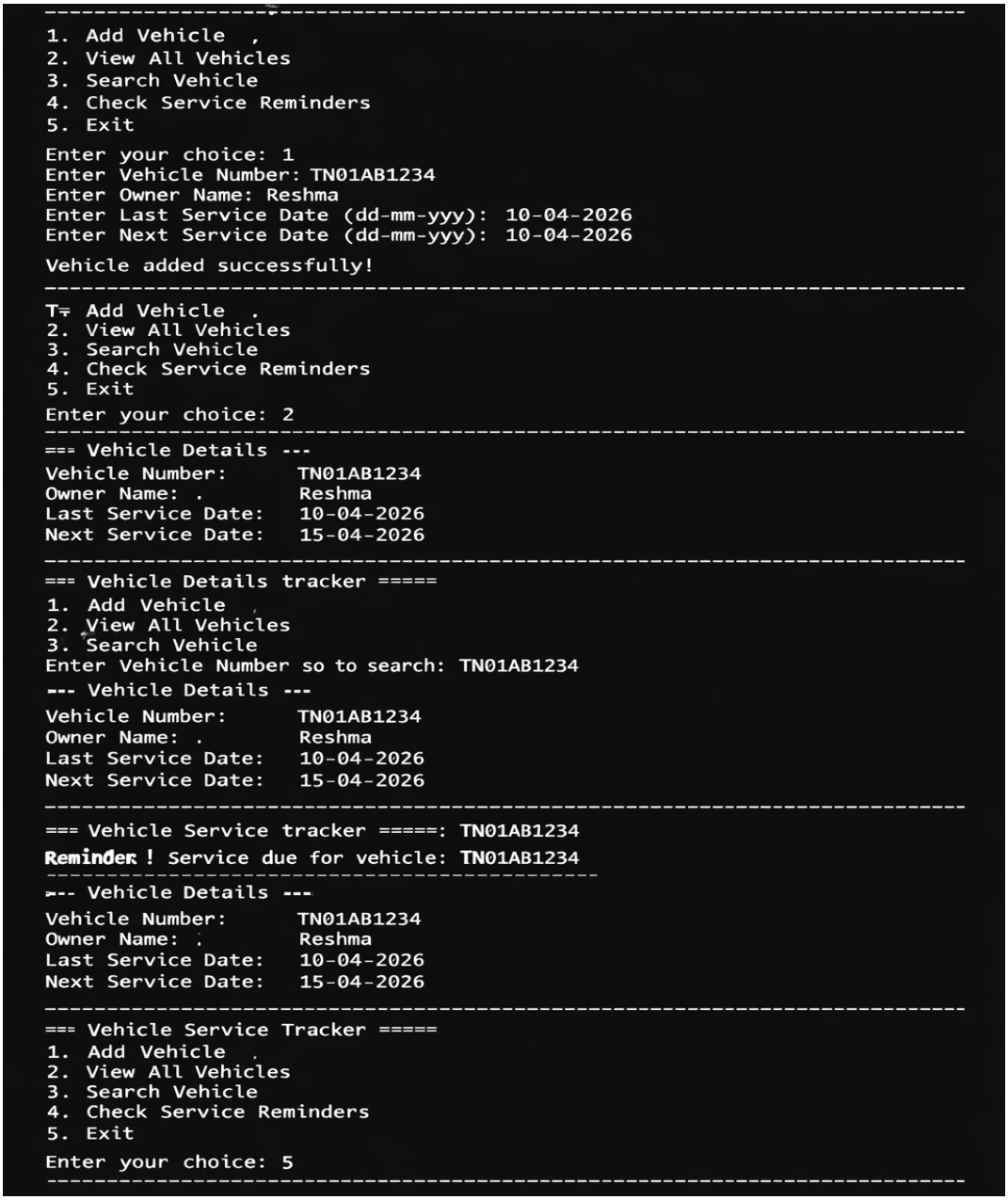
    static void checkReminders(Scanner sc) {
        System.out.print("Enter today's date (dd-mm-yyyy): ");
        String today = sc.nextLine();

        boolean found = false;

        for (Vehicle v : vehicles) {
            if (v.nextServiceDate.equals(today)) {
                System.out.println("\nReminder! Service due for vehicle: " + v.vehicleNumber);
                v.display();
                found = true;
            }
        }

        if (!found) {
            System.out.println("No service reminders for today.");
        }
    }
}
```

5. INPUT AND OUTPUT SCREENS



6. DISCUSSION & CONCLUSION

7.1 Discussion

- The developed system provides a practical solution for managing vehicle maintenance activities. It simplifies the process of tracking service schedules and reduces the chances of missing important maintenance tasks. The application works efficiently by storing vehicle data and calculating service reminders based on user input.

- The system is designed with multiple modules such as vehicle entry, reminder generation, and record management. Each module performs a specific function, ensuring smooth operation of the application. The reminder feature plays a key role by notifying users about upcoming or overdue services.

7.2 Conclusion

The Vehicle Service Reminder and Maintenance Tracker is a useful application that helps users maintain their vehicles effectively. It provides an easy way to store service details and receive reminders for maintenance activities. The system reduces manual work and ensures proper record management.

By using this application, users can improve vehicle performance, reduce maintenance costs, and avoid unexpected breakdowns. The project demonstrates how simple software solutions can solve real-life problems efficiently.

In conclusion, this system is a reliable and efficient tool for vehicle service management. With further enhancements, it can be developed into a more advanced and user-friendly application suitable for wider use.

7. REFERENCES

Books

- **Java Programming by Herbert Schildt**
- **Object-Oriented Programming with Java**

Websites

- http://www.w3schools.com
- http://www.geeksforgeeks.org
- http://www.javatpoint.com